

Name _____

Roll No _____

INDEX**VBDS6P4: Applied Business Analytics Practical**

No.	Title/Aim	Date	Student Sign	Grade	Faculty Sign
1	Introduction to Business Analytics: Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset. Analyze customer data to identify trends and patterns that can be used for business decision-making.				
2	Describing the Distribution of a Variable: Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics				
3	Finding Relationships Among Variables: Use a dataset with multiple variables and perform correlation analysis to determine the strength and direction of relationships between pairs of variables. Apply regression analysis to identify the relationship between an independent variable (e.g., advertising expenditure) and a dependent variable (e.g., sales revenue).				
4	Business Intelligence Tools for Data Analysis: Utilize a business intelligence tool (e.g., Tableau, Power BI) to extract insights from a dataset and create interactive visualizations for effective data analysis				
5	Probability and Probability Distributions: Simulate a probability experiment (e.g., rolling dice) using programming and calculate the probabilities of different outcomes. Generate random numbers from various probability distributions (normal, uniform, exponential) and analyze their properties				
6	Decision Making under Uncertainty: Develop a decision tree model to make business decisions considering uncertainties and associated probabilities at each decision point. Apply the concept of expected value to evaluate different decision alternatives and select the optimal one				
7	Sampling and Sampling Distributions: Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.				

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	Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters				
8	Confidence Interval Estimation: Calculate confidence intervals for population means or proportions using sample data and interpret the results in a business context. Apply bootstrapping techniques to estimate confidence intervals for non- parametric statistics				
9	Hypothesis Testing: Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the results. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions.				
10	Regression Analysis and Time Series Analysis: Develop a regression model to predict future sales based on historical data, assess model performance, and interpret the significance of predictor variables. Apply time series analysis techniques (e.g., ARIMA, exponential smoothing) to forecast future demand for a product or service, and evaluate the accuracy of the forecasts.				
11	Optimization Modeling and Simulation Modeling: Formulate an optimization model (e.g., linear programming, integer programming) to solve a real-world business problem and analyze the optimal solution. Use simulation modeling to evaluate different business scenarios, such as capacity planning, inventory management, or pricing strategies, and assess their impact on performance metrics.				
12	Analysis of Variance and Experimental Design: Design and conduct an experiment to study the effects of different factors on a specific response variable, analyze the results using analysis of variance (ANOVA), and draw conclusions. Implement a factorial experiment and analyze the main effects and interaction effects of factors using statistical techniques.				
PBL	Create Customer Segmentation & Retention Dashboard				
PBL	Create Student Educational Analysis Dashboard				
Newly Added Expt	Customer Segmentation using Clustering				

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Newly Added Expt	Market Basket Analysis (Association Rules)				



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CERTIFICATE

This is to certify that the practicals in the subject of
VBDS6P4: Applied Business Analytics Practical
 have been completed satisfactorily by

Mr./Ms. _____ Roll no. _____ in
 partial fulfilment of BSc DS degree (Sem VI) examination for Academic Year 2025 – 26

 Practical Instructor

 Program Coordinator

 Internal Examiner

 External Examiner

 PRINCIPAL

Date: / / 2026

Date: / / 2026

College Seal